

2. Advanced Testing Types & When to Use Them



Page is about?

This page will:

- Bring clarity to all types of testing
- Standardize language across teams
- Help product/QA/devs ask for the *right type* of test
- Align execution with **business risk & infra behavior**

1. Overview of Performance Test Types

Test Type	Purpose	Duration	Triggers
Load Test	Validate system under expected normal conditions	30–60 mins	Before every major release
Stress Test	Determine system breaking point under extreme load	Until crash/failure	For scalability, capacity planning
Spike Test	Test reaction to sudden, high traffic surge	15–30 mins	For marketing, flash sales, rollout events
Soak Test	Check stability over long durations	4–12 hours	For memory leaks, CPU/thread saturation
Endurance Test	Continuous test to simulate production	12–72 hours	For production mimic, batch jobs, DB perf

2. When to Use Which Test?

- Goal: Confirm that response time, TPS, and error rates are within SLA under typical production load
- Run before major releases (e.g. Q2, Q3)
- Example: 300 users, 60 min, 3 TPS

Ideal for: Dashboard, Enrollment APIs, Plan Lookup

- Goal: Identify where system breaks (memory, threads, DB locks)
- Increase load beyond known thresholds
- Example: 100 800 users, spike every 2 mins

Ideal for: Sizing analysis, DB index tuning, Redis burst

- Goal: Analyze reaction to sudden load increase & quick return
- Example: 50 users 250 50 within 10 mins

Ideal for: Login APIs, Dashboard traffic, 9am/5pm loads

- Goal: Detect memory leaks, CPU rises, session issues
- Example: 100 users for 6 hours, consistent activity

Ideal for: CSR daily flows, back-office jobs

- Goal: Mimic 24–72hr continuous usage including nightly jobs
- Requires stable environment + monitoring

Ideal for: Production simulation, payroll, batch jobs

3. YAML & JMeter Examples

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execution:

- concurrency: 300
hold-for: 60m
ramp-up: 10m
scenario: login_api

—
execution:

- concurrency: 50
ramp-up: 2m

- concurrency: 250
hold-for: 5m
scenario: login_api
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- Threads: 500
- Ramp-Up: 5 min
- Duration: 45 min
- Loop Controller: Forever/Timed

4. Reporting & Metrics by Test Type

Test Type	Key Metrics
Load	Avg/90th/95th Response Time, TPS, Error %, CPU/RAM
Stress	Max Capacity, Fail Point, Heap Dump Analysis
Spike	Recovery Time, GC Lag, Load Balancer behavior
Soak	Thread/Memory Growth Over Time, Leaks
Endurance	Data Drift, Schedulers, Long-Run Trends

5. Common Use Cases

Use Case	Test Type	Notes
Login at scale	Spike	Validate failover/resilience
CSR operations	Soak	High user, steady load
529 Enrollment peak	Load + Spike	Critical test before Q2
Platform tuning	Stress	Before scaling infra
Nightly job overlap	Endurance	Need 48–72hr view

6. Best Practices & Pro Tips

- Always baseline with Load Test before other types
- Never run Stress/Spike on shared QA without DevOps alert
- Label your YAML files clearly (dashboard_spike.yaml)
- Tag your reports for traceability (perf-spike-q2-529.yaml)
- Include heap graphs, CPU, and GC logs in Stress/Soak reports